An Aiming Device for Steerable Needles

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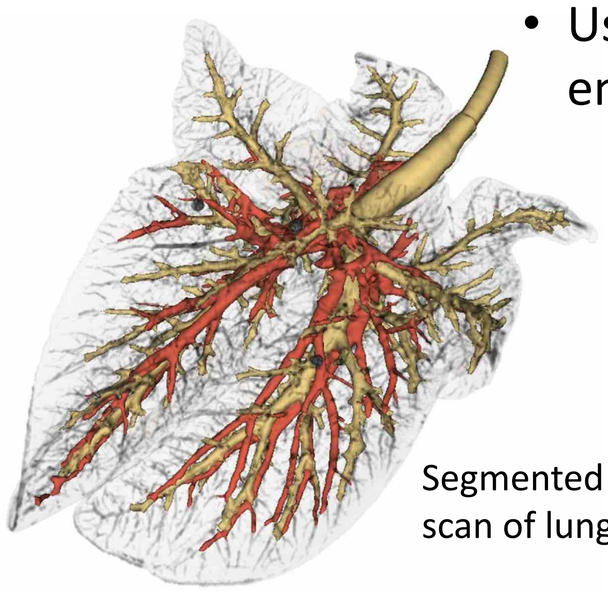
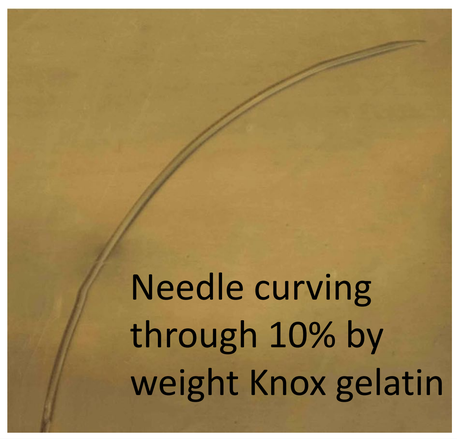
Steerable Needles

- Capable of obstacle avoidance and curving through soft tissue
- Useful for obstacle dense environments such as lungs



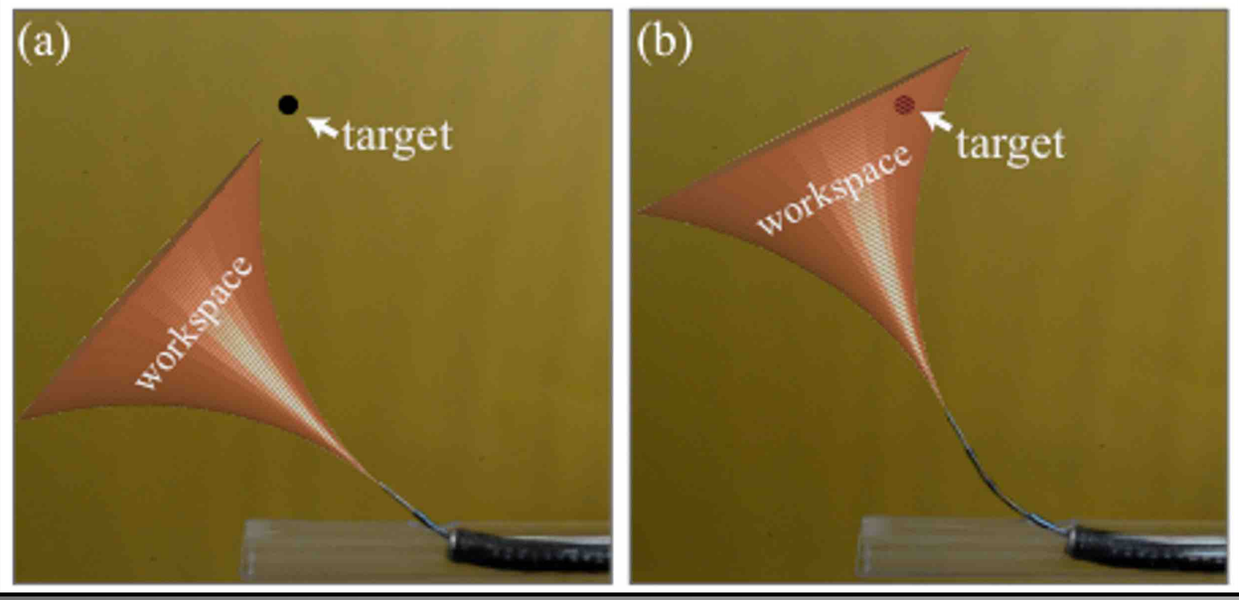
Segmented CT scan of lungs



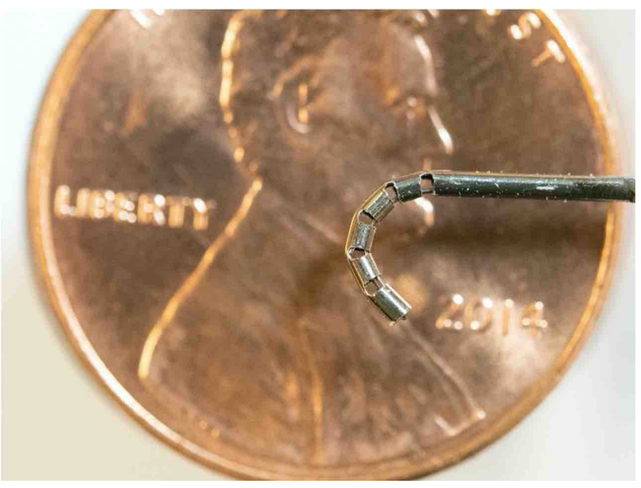
Needle curving through 10% by weight Knox gelatin

Problem Statement

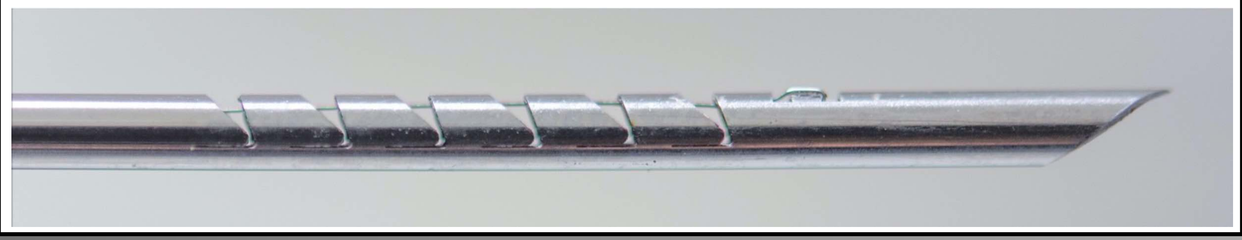
- Steerable needles have a defined trumpet-shaped workspace
- Initial launch pose determines needle's ability to reach targets



New Aiming Device

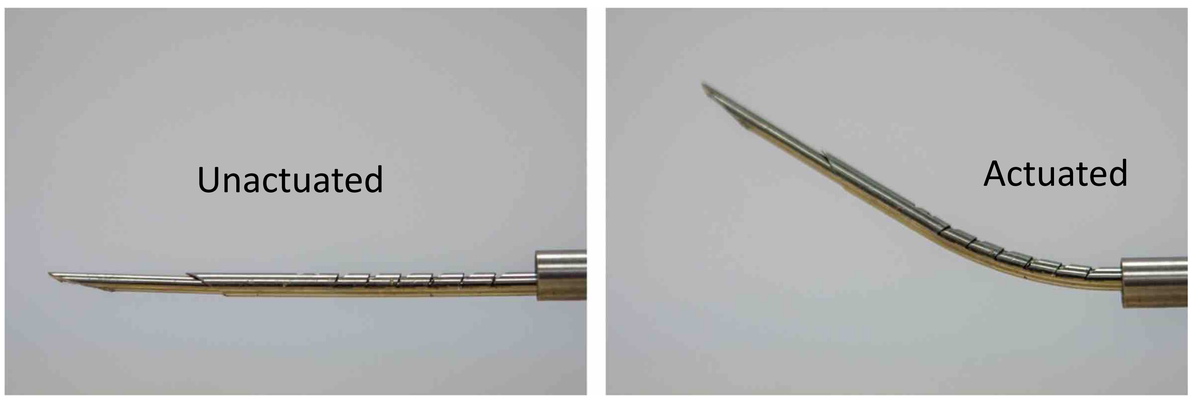


- Notched tubes have previously been used as manipulators [1]
- Adapted concept with new notch profile [2] to use as aiming device

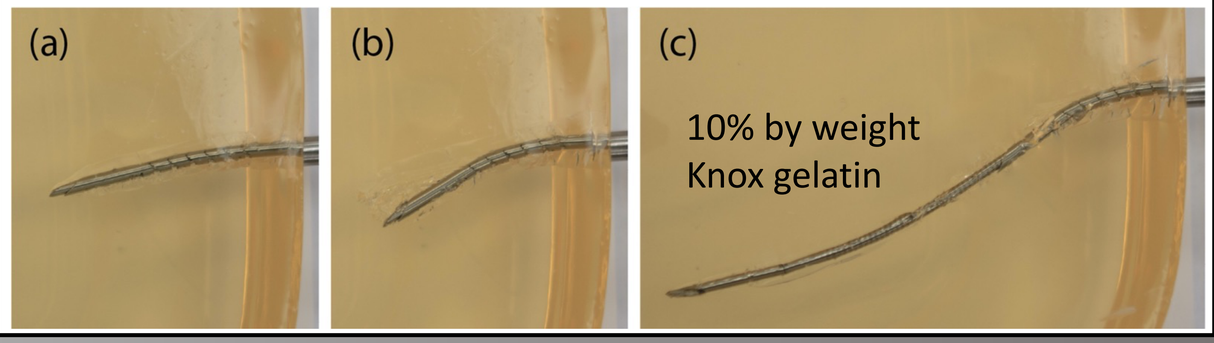


Actuation Demonstration

- Tendon wire located at tip of aiming device applies tension to actuate the device

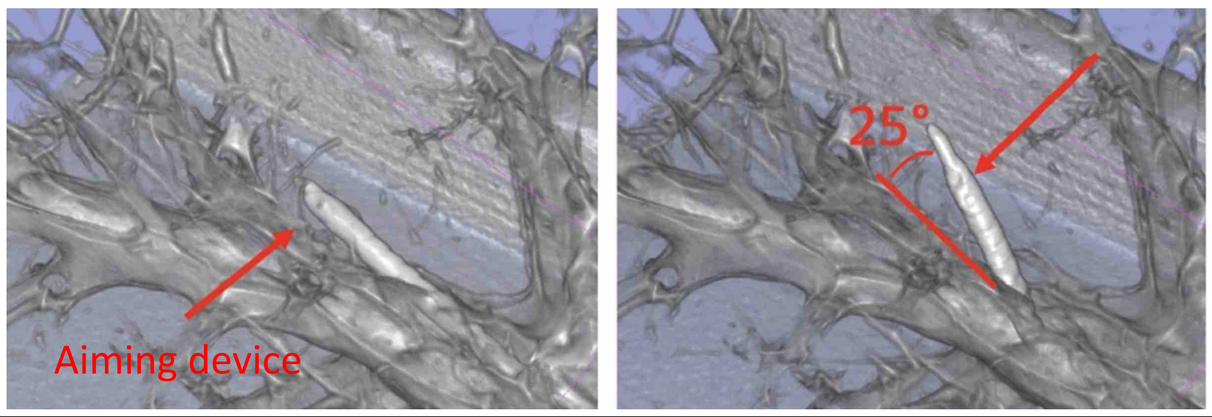


- Workflow:
  a) Aiming device is inserted in tissue
  b) Tendon wire is pulled to desired tension
  c) Needle is steered to target



Clinical Application

- Aiming device tested in ex-vivo porcine lung using robotic system described in [3]
- Needle is deployed bronchoscopically and enters lung tissue from the airways
- CT scan below shows device re-orienting the initial launch pose once needle is in lung tissue



References

[1] P. J. Swaney, P. A. York, H. B. Gilbert, J. Burgner-Kahrs, and R. J. Webster III, "Design, Fabrication, and Testing of a Needle-Sized Wrist for Surgical Instruments," Journal of Medical Devices, vol. 11, no. 1, p. 014501, 2017.

[2] K. W. Eastwood, P. Francis, H. Azimian, A. Swarup, T. Looi, J. M. Drake, and H. E. Naguib, "Design of a Contact-Aided Compliant Notched-Tube Joint for Surgical Manipulation in Confined Workspaces," Journal of Mechanisms and Robotics, vol. 10, no. 1, p. 015001, 2018.

[3] S. R. Amack, M. F. Rox, J. E. Mitchell, T. E. Ertop, M. Emerson, A. Kuntz, F. Maldonado, J. Akulian, J. B. Gafford, R. Alterovitz, and R. J. Webster III, "Design and Control of a Compact, Modular Robot for Transbronchial Lung Biopsy," in SPIE Medical Imaging, vol. 10951, 2019.

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